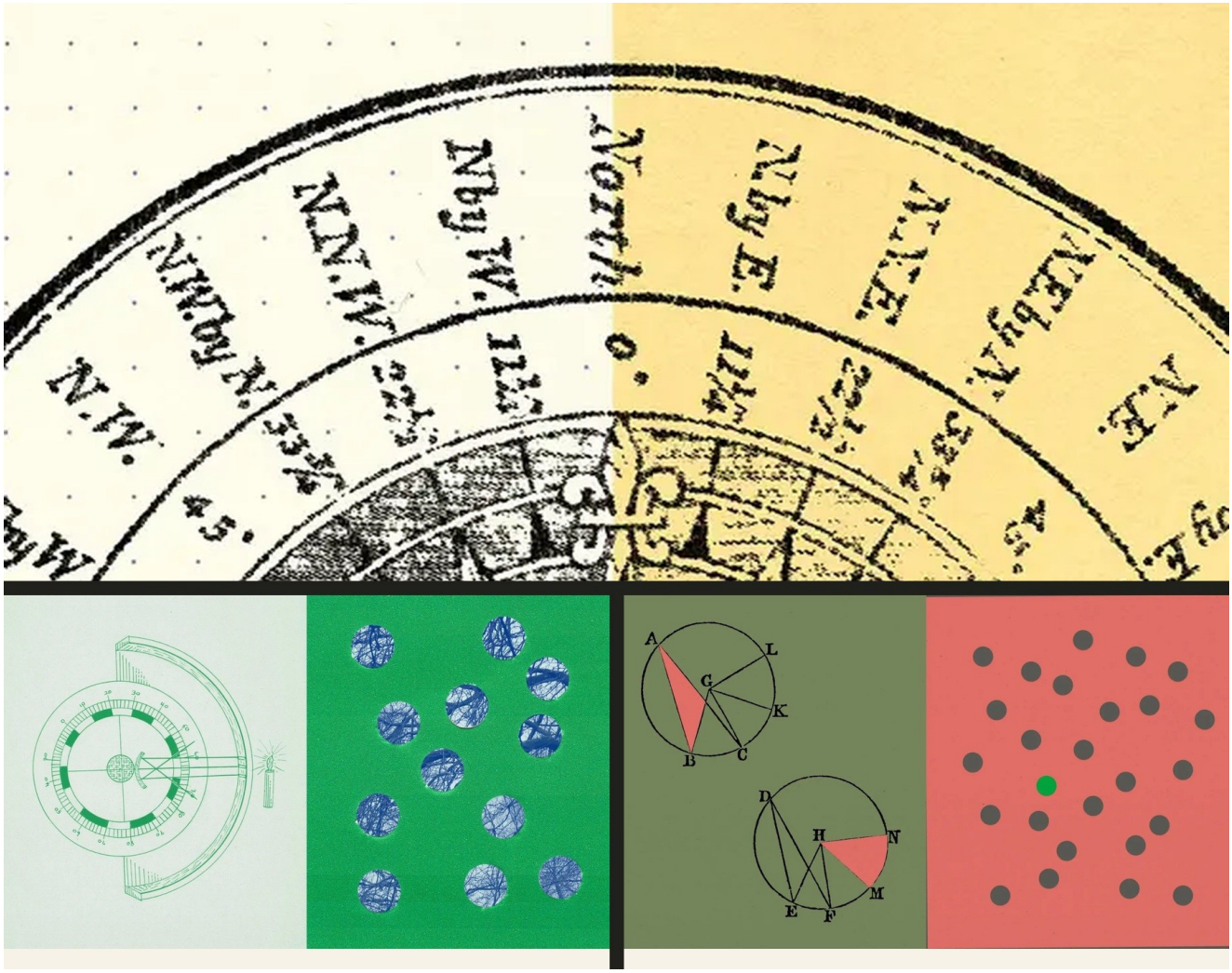



OBLIVION / MODEL SYSTEMS REPORT

# GPT-OSS 20B & 120B

## Architecture, reasoning behaviour, evaluation and deployment engineering

A critical engineering assessment for the Oblivion team

ENGLISH EDITION | 26 October 2026 | Research / 01

01 / OBLIVION RESEARCH

# Executive assessment

Two operating points - one engineering decision framework

gpt-oss is a family of open-weight reasoning models designed for developers who want to inspect, host, adapt and operate model weights within infrastructure they control. Its significance is not captured by parameter count alone. The family combines sparse Mixture-of-Experts computation, adjustable reasoning effort, Harmony-formatted tool interaction and Apache 2.0 licensing, subject to the published usage policy.

The 20B and 120B variants should be treated as different operating points. The smaller model is easier to host and can be highly competitive on bounded reasoning tasks. The larger model offers more total capacity and stronger results on several knowledge, coding and tool-use evaluations, but demands substantially more memory and compute. Neither is the universal winner.

| Measure | gpt-oss-20b | gpt-oss-120b |
|---|---|---|
| Total parameters | 20.91B | 116.83B |
| Active parameters / token | 3.61B | 5.13B |
| Checkpoint size | 12.8 GiB | 60.8 GiB |
| Selected experts / token | 4 | 4 |

- Open-weight does not mean fully reproducible open-source training: the public release does not disclose the complete pretraining corpus and full training pipeline.
- Active parameters describe selected computation, not the amount of weight data that must be stored.
- gpt-oss is not served through ChatGPT or the OpenAI API; deployment responsibility sits with the operator or hosting provider.

Engineering verdict: use 20B as the first candidate for constrained local workflows; consider 120B for difficult knowledge, multi-step coding and tasks where Oblivion measurements prove a quality gain worth the infrastructure cost. Route by task and evidence, not by brand or parameter count.

Scope update: OpenAI also lists gpt-oss-safeguard-20b and -120b as research-preview safety-reasoning models. They are policy-classification derivatives, not replacements for the general-purpose 20B/120B models discussed here.

02 / OBLIVION RESEARCH

# Architecture and sparse computation

Why active parameters are not the same as model size

Both models are autoregressive Transformer language models built around Mixture-of-Experts (MoE) blocks. For each token, a learned router selects four experts from a larger pool. Only the selected experts participate in that token's expert computation, while shared layers and attention remain part of the overall forward pass.

This sparse routing improves the ratio between total model capacity and per-token expert computation. It does not shrink the checkpoint to the active-parameter count: all expert weights must be available in memory, streamed or offloaded by the runtime.

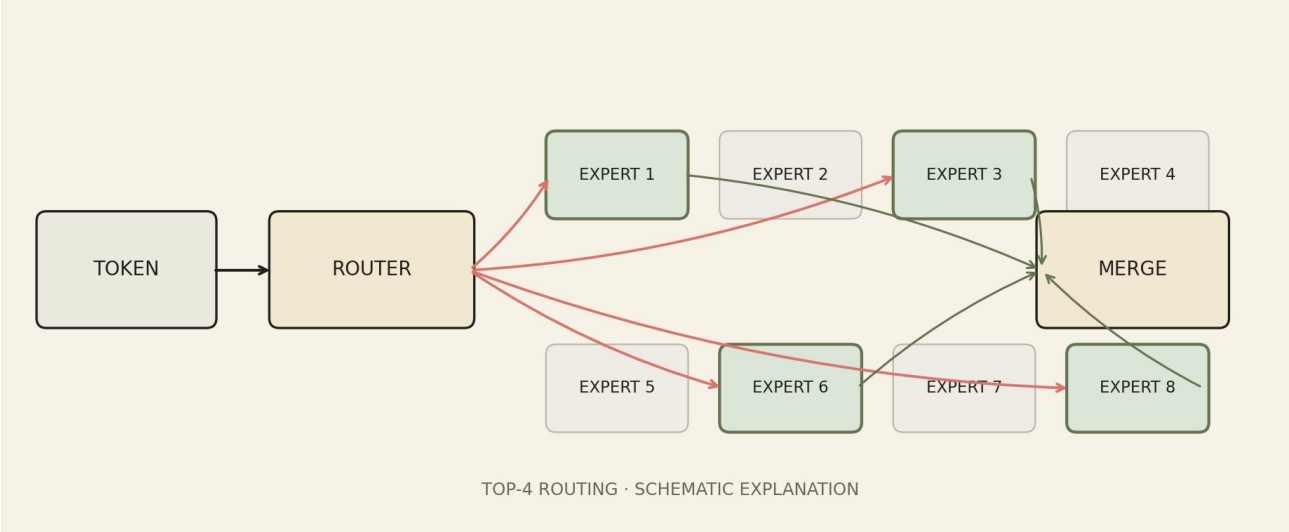



*Figure 1. Conceptual top-4 MoE routing. The diagram is explanatory, not a reproduction of the internal block graph.*

| Architecture item | 20B | 120B |
|---|---|---|
| Transformer layers | 24 | 36 |
| Experts per MoE block | 32 | 128 |
| Experts selected per token | 4 | 4 |
| Query heads | 64 | 64 |
| Key/value heads | 8 | 8 |
| Attention head dimension | 64 | 64 |

- Attention alternates local banded-window and dense patterns; the local window bandwidth is 128 tokens.
- Grouped-query attention uses 64 query heads and 8 key/value heads.
- RoPE and YaRN support a reported context length up to 131,072 tokens in dense layers.
- The reported stack includes RMSNorm, Pre-LN, SwiGLU and MXFP4 post-training quantization for MoE weights.

Interpretation: 3.61B and 5.13B are useful measures of per-token active computation. They are not reliable estimates of RAM/VRAM requirements. For memory planning, start with checkpoint size, then add KV cache, runtime buffers, batching and operating-system overhead.

03 / OBLIVION RESEARCH

# Evaluation: what the published scores say

A comparative reading of high-reasoning results

The table below reproduces selected high-reasoning values from OpenAI's Model Card. Tool-assisted and no-tool rows are different evaluation settings. Scores from unrelated benchmarks are not directly comparable, and vendor-reported scores are not independent replications by Oblivion.

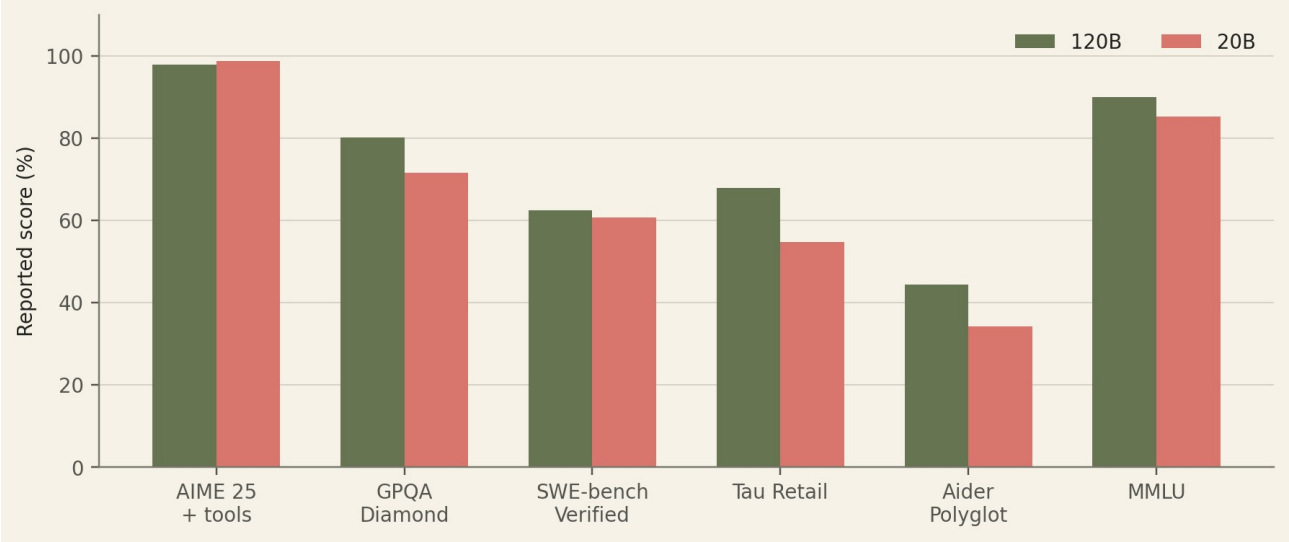



*Figure 2. Selected scores reproduced from OpenAI's published Model Card. The AIME 2025 tool-assisted result is the notable row where 20B is higher.*

| Benchmark / setting | 120B | 20B | Higher score |
|---|---|---|---|
| AIME 2025 · tools | 97.9 | 98.7 | 20B |
| GPQA Diamond · no tools | 80.1 | 71.5 | 120B |
| SWE-bench Verified | 62.4 | 60.7 | 120B |
| τ-Bench Retail | 67.8 | 54.8 | 120B |
| Aider Polyglot | 44.4 | 34.2 | 120B |
| MMLU | 90.0 | 85.3 | 120B |

- 120B has larger reported advantages on GPQA Diamond, τ-Bench Retail and Aider Polyglot.
- 20B slightly exceeds 120B on the published AIME 2025 tool-assisted row; this is task-specific, not evidence of a general reversal.
- Codeforces Elo is reported as 2,622 vs 2,516 with tools and 2,463 vs 2,230 without tools (120B vs 20B).

Do not compress this into a single leaderboard. Model choice should be based on the Oblivion task mix: coding repository changes, local extraction, multilingual support, tool reliability and latency may rank the models differently.

04 / OBLIVION RESEARCH

# Reasoning, Harmony and tool use

A model that can call tools is not yet a dependable agent

Reasoning effort is configurable as low, medium or high. Greater effort can improve results on difficult problems, but it usually increases generated tokens, latency and compute. OpenAI reports that gpt-oss-20b averaged more than 20,000 chain-of-thought tokens per AIME problem in a high-reasoning setting. This is a warning about cost and latency, not a target budget for production.

The models use Harmony, a serialized conversation format that carries message boundaries, roles, channels and control-flow markers. Tool calls are assistant messages directed to a tool; tool responses return under the tool's name. An integration must parse control flow correctly rather than treating the entire stream as plain text.

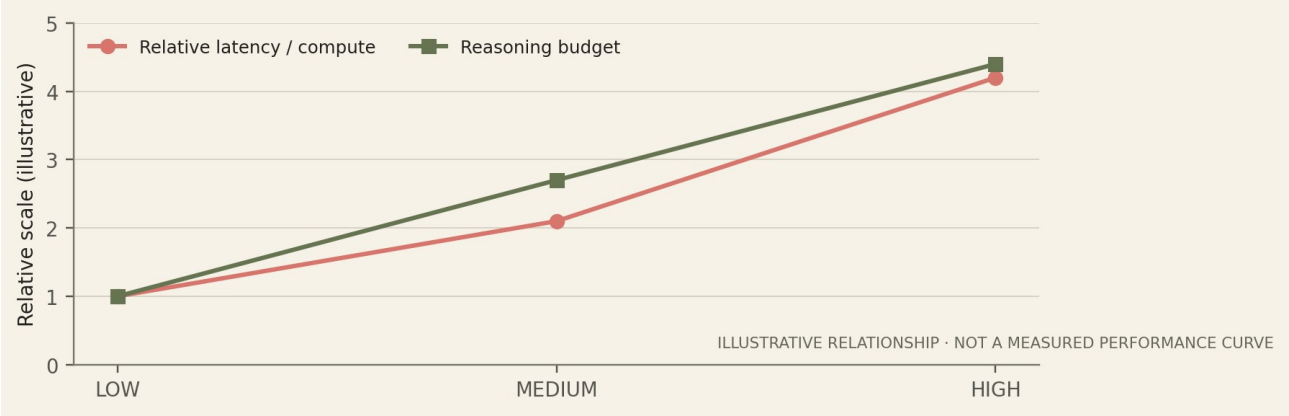



*Figure 3. Qualitative relationship between reasoning budget and compute. This is an explanatory illustration, not a measured curve.*

| Control surface | Engineering implication |
|---|---|
| Reasoning effort | Set per task; cap tokens and time. |
| Harmony parser | Validate roles, channels, recipients and hand-off markers. |
| Tool schemas | Validate arguments and reject unknown functions. |
| State machine | Track tool calls, retries, timeouts and completion. |

- Keep user data, tool output and developer instructions in distinct trust domains.
- Treat retrieved documents and web pages as untrusted content, including any embedded instructions.
- Use bounded retries, explicit stop conditions and a maximum tool-call budget.

Never expose raw internal reasoning as a guaranteed explanation. OpenAI warns that raw CoT can contain hallucinated or policy-inconsistent content. Present concise rationales, tool outputs, citations and test evidence instead.

05 / OBLIVION RESEARCH

# Inference, memory and deployment

Checkpoint size is only the first line of the resource budget

OpenAI reports checkpoint sizes of 12.8 GiB for 20B and 60.8 GiB for 120B. These are weight-storage figures, not complete runtime requirements. Actual deployment also depends on KV cache, context length, concurrent sequences, allocator fragmentation, quantization format, offload policy and runtime implementation.

A 131K context ceiling does not mean that every prompt should use the full context. Long contexts increase KV-cache pressure and can reduce throughput. A model that technically loads may still be too slow, unstable or expensive for a product workflow.

| Resource factor | What to measure |
|---|---|
| Weights | Resident VRAM/RAM; format; offload ratio. |
| KV cache | Peak memory by context length and concurrency. |
| Latency | Time to first token; p50/p95 completion time. |
| Throughput | Input/output tokens per second at target batch size. |
| Reliability | OOM rate, restarts, malformed tool calls. |
| Cost / energy | GPU time and energy per accepted task. |

- OpenAI positions 120B for a single 80 GB GPU; confirm exact runtime and context conditions before treating that as an operational guarantee.
- For 20B, do not infer smooth execution from a nominal 16 GB target alone; check whether the capacity refers to VRAM, unified memory or system RAM.
- Record GPU model, driver, runtime version, quantization, batch size, context, prompt length and generation budget for every benchmark.

Deployment rule: benchmark the full product path, not just model loading. Measure successful task completion per unit of latency, memory, energy and operator intervention.

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# Safety, reliability and openness boundaries

The weights do not enforce product policy by themselves

OpenAI’s Model Card notes that open-weight deployment shifts substantial responsibility to downstream developers. Fine-tuning and system composition can alter behaviour, and the publisher cannot centrally enforce the safeguards of every derivative deployment. Apache 2.0 supports broad use and modification, but the published usage policy remains relevant.

The Model Card reports that the models generally underperform o4-mini on instruction-hierarchy evaluations and on SimpleQA/PersonQA factuality tests. The report also cautions that raw internal reasoning may contain hallucinations or policy-inconsistent language. These limits matter more when a model has tools or access to local files.

| Risk | Required Oblivion control |
|---|---|
| Prompt injection | Treat retrieved content as data, never as policy; isolate tool instructions. |
| Over-privileged tools | Allowlist capabilities; read-only defaults; explicit approval for writes. |
| Factual errors | Require source citations, retrieval checks and claim verification. |
| Unsafe code execution | Sandbox, resource limits, no unrestricted shell or secrets. |
| Silent failures | Structured logs, event IDs, retries with limits and alerting. |

- OpenAI reports parity with o4-mini on some StrongREJECT jailbreak evaluations, but weaker resistance to user attempts to override system instructions.
- The default 120B did not reach the report’s indicative “High” thresholds in tracked Preparedness categories; this is a bounded evaluation result, not a universal safety guarantee.
- Open weights can be modified after release. A product must enforce its own permissions, monitoring, moderation and audit trail.

Security position: safety is a system property. It depends on the model, prompts, runtime, tools, permissions, data sources, human review and monitoring—not on the model checkpoint alone.

07 / OBLIVION RESEARCH

# Oblivion reference architecture

Keep the model replaceable; keep policy outside the model

Oblivion should treat gpt-oss as an interchangeable reasoning backend. A separate orchestration layer should own authentication, user permissions, model routing, context assembly, tool authorization, execution isolation, evidence checking and audit events. This architecture remains valid if the underlying model changes.

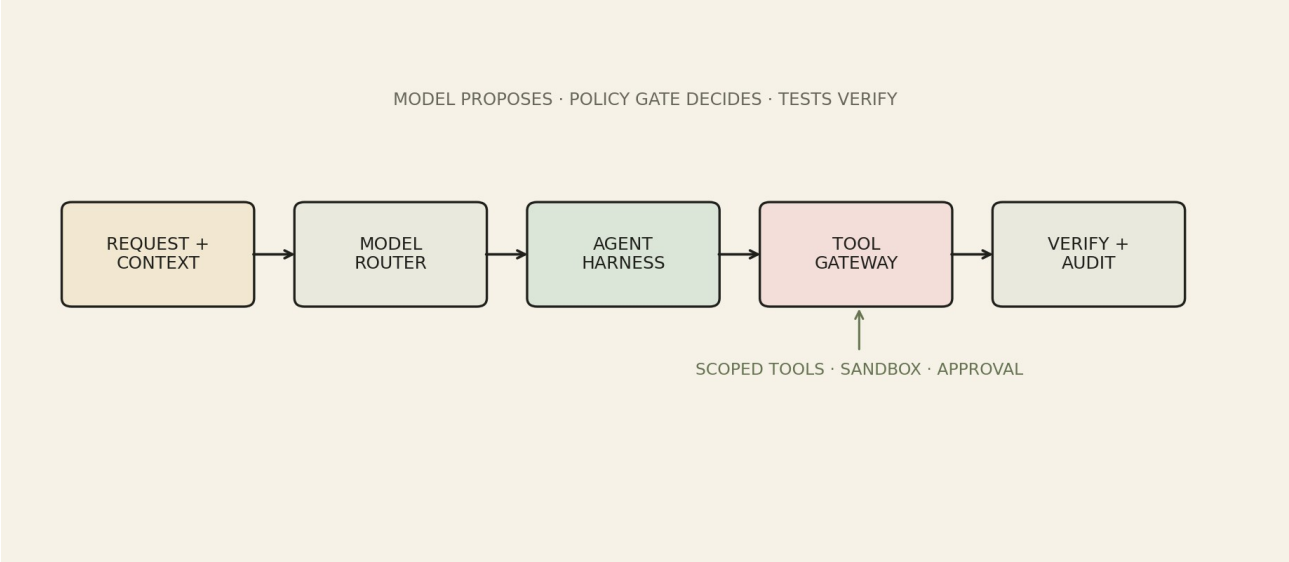



*Figure 4. Suggested separation of concerns for an Oblivion agent harness.*

| Layer | Responsibility |
|---|---|
| Identity and policy | User/session identity, tenant boundaries, role checks. |
| Model router | Select 20B/120B using task class, risk, budget and measured quality. |
| Agent harness | Harmony adapter, state machine, token/time budgets, bounded retries. |
| Tool gateway | Allowlist, schema validation, sandbox, approval gate. |
| Verifier | Tests, citations, diff review, output constraints. |
| Observability | Immutable audit events, metrics, trace IDs and incident review. |

- 20B candidate role: extraction, summarization, first-pass code drafts, bounded local analysis and low-risk tasks with validation.
- 120B candidate role: difficult planning, complex debugging, deep code review and tasks where internal evaluation demonstrates meaningful uplift.
- Require human approval for deletion, publishing, external messaging, credential changes, repository merges and other state-changing actions.

The model may propose an operation; the tool gateway decides whether that operation is authorized. A model score is never a permission grant.

08 / OBLIVION RESEARCH

# Three-pass critical review

Claims challenged, evidence checked, design revised

Pass 1 - Challenge parameter-count assumptions. The first draft implicitly treated 20B as “small” and 120B as “better”. Both are incomplete. The 20B checkpoint is 12.8 GiB before runtime overhead; 20B also exceeds 120B on one AIME 2025 tool-assisted row while trailing on several knowledge and agentic tasks. Revision: describe task-specific operating points, not a universal ranking.

Pass 2 - Challenge benchmark interpretation. Published scores do not establish identical prompt, harness, repetition or hardware conditions for every task. High reasoning raises compute use. Revision: label every score with its setting, record uncertainty where available and reproduce results on an Oblivion task suite before selecting a production default.

Pass 3 - Challenge safety-by-prompt. A prompt cannot reliably enforce policy when weights are modifiable and tool execution is external. Revision: put permission checks, sandboxing, output validation and human approval in code outside the model.

- Build a minimum 100-task evaluation set from anonymized Oblivion workflows.
- Test Persian, English, German, Japanese and Korean separately; do not extrapolate from English-only scores.
- Compare identical prompts, runtime, tools, budgets and hardware; report low/medium/high reasoning separately.
- Measure task pass rate, unit tests, hallucination, injection resistance, tool misuse, p50/p95 latency, peak memory and energy per accepted task.
- Keep the deployment read-only by default; require approval for side effects and preserve complete audit logs.

Final position: 20B is a resource-conscious candidate with meaningful reasoning capability; 120B offers greater capacity and stronger results on several demanding evaluations. Neither is a complete product. Oblivion’s value must come from orchestration, verification, security and measured reliability.

09 / OBLIVION RESEARCH

# Methodology and primary references

Evidence provenance, reproducibility and limits

This report was rewritten as an engineering assessment for the Oblivion team. Architecture and benchmark figures are attributed to OpenAI's published gpt-oss Model Card. Current hosting and Safeguard-family notes are based on OpenAI Help Center guidance available on 26 October 2026. Harmony behaviour is cross-checked against OpenAI's public format documentation.

No benchmark in this report was independently run by Oblivion. Reported values can vary with harness version, prompt design, reasoning effort, tool implementation, quantization, hardware, decoding settings and evaluation protocol. The diagrams are explanatory editorial illustrations; they are not source diagrams from OpenAI.


- [1] OpenAI, gpt-oss-120b & gpt-oss-20b Model Card, 5 August 2025. https://deploymentsafety.openai.com/gpt-oss/a2
- [2] OpenAI, Introducing gpt-oss, 5 August 2025. https://openai.com/index/introducing-gpt-oss/
- [3] OpenAI Help Center, OpenAI open-weight models; current guidance accessed 26 October 2026. https://help.openai.com/
- [4] OpenAI, Harmony format documentation and public repository. https://github.com/openai/harmony
- [5] OpenAI Cookbook, Harmony format. https://cookbook.openai.com/articles/openai-harmony
- [6] Hugging Face, official OpenAI gpt-oss model repositories. https://huggingface.co/openai


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